# Textual Anomaly Detection: A Systematic Evaluation of Representations and Algorithms

Panagiotis Soustas[0009-0002-3741-877X]
and Matthew Edwards[0000-0001-8099-0646]

University of Bristol
{ki22640,matthew.john.edwards}@bristol.ac.uk

**Abstract.** This study evaluates scalable, unsupervised methods for detecting malicious online content by benchmarking transformer-based architectures across diverse linguistic contexts. We demonstrate that the semantic resolution of the embedding layer is the critical determinant of performance. Our results identify that large language models coupled with manifold learning achieve superior anomaly separation in knowledge-intensive domains, significantly outperforming traditional BERT-based pipelines. Additionally, we reveal a topological dichotomy in detection strategies: Contrastive autoencoders offer robust stability in structured environments, whereas few-shot deviation learning (FATE) is essential for high-entropy, dynamic contexts such as politics and sports. These findings propose a shift toward context-aware architectures capable of adapting to the complex semantic landscape of modern web content.

**Keywords:** Linguistic Anomaly Detection, Unsupervised Learning, Large Language Models, Contrastive Learning

## 1 Introduction

The increasing complexity of modern text data has rendered static, rule-based filtration systems insufficient for capturing sophisticated, evasive linguistic patterns. This paper addresses this limitation by proposing a methodological shift: rather than categorizing specific types of online harm, we frame the detection challenge as one of linguistic anomaly detection—identifying inputs that deviate from the semantic topology of a learned context. The core challenge, therefore, is to identify text that is contextually anomalous. The effectiveness of any such detection system depends critically on two components, text representation and anomaly detection.

Within natural language processing, anomaly detection is fundamental for identifying unusual text instances, with application such as spam detection [19], misinformation [13] or anomalous web content, like reviews or posts [22]. Textual anomaly detection poses some challenges due to the nature of language, which is unstructured, with anomalies emerging through subtle irregularities in semantics, pragmatics, and context [25]. Additionally, as with most anomaly detection domains, anomalous instances of text are infrequent, making acquiring accurately labelled anomalous samples difficult.

This study provides a systematic and empirical evaluation of different combinations of NLP feature representation techniques and anomaly detection algorithms. Our goal is to answer a critical question for the field of online safety: Which methodologies are most effective at identifying contextual language anomalies indicative of malicious content? By systematically evaluating the interplay between feature extraction and anomaly detection, this paper offers three primary contributions:

- A comprehensive benchmark of pipelines for textual anomaly detection.
- Empirical guidance on which NLP representations are best suited for capturing specific types of contextual anomalies.
- An analysis of whether combinatorial pipelines offer advantages over already existing stand-alone models.

To achieve this, we evaluate a selection of robust language models in combination with a range of anomaly detection algorithms, comparing the performance of each combination on benchmark datasets drawn from previous literature [22,18,5]. These datasets are tailored for evaluating the performance of text-based anomaly detection models in a realistic, heterogenous environment. The key aspect of the methodology is the comparative analysis. This requires a design where the different combinations of methods are not only evaluated separately but also in combination, to highlight their relative strengths and weaknesses. This provides a foundational roadmap for building more effective, scalable and linguistically-aware systems to identify contextual anomalies and irregularities across diverse environments.

## 2 Related Work

Modern textual anomaly detection research presents two-stage solutions [25,26] in which textual data is first converted to a structured numerical format as part of feature engineering, before an anomaly detection algorithm is applied to identify outliers within these representations. Previous research lacks consensus on the optimal pairing between embeddings and detection algorithms.

While the earlier works [22,18,5,10] focused on contextualized embeddings from models like BERT [11] the emergence of LLMs took the spotlight in generating text representations. However, the latest comparisons coupled the LLMs with a set of shallow anomaly detection algorithms [25,26]. Despite the observation that embedding quality governs anomaly detection efficacy, a comprehensive systematic evaluation has yet to be conducted that tests these pairings across a diverse suite of representation and algorithm types. Thus the existing benchmarks suffer from limited comparative analysis. We attempt to fill in this gap by providing a comprehensive evaluation across diverse text representation strategies and a wide array of both shallow and deep anomaly detection algorithms.

## 3 Methodology

We evaluate the performance of different combinations of NLP feature representations and anomaly detection models through a structured, multi-stage experiment. The primary objective is to determine which combinations of text embedding strategies and anomaly detection algorithms are most effective for identifying linguistic anomalies, with a particular focus on the performance enhancements offered by a contrastive learning framework.

### 3.1 Datasets

We use two standard benchmarks widely adopted in NLP research: the 20 Newsgroups dataset [3] and the AG News corpus [28]. These datasets provide a diverse range of linguistic contexts, varying from highly technical jargon to conversational text, enabling assessment of model performance across different domains.

The 20 Newsgroups dataset[1] is a classic benchmark for text classification and clustering, originally collected by Ken Lang. It comprises 18,846 documents, partitioned nearly evenly across 20 distinct newsgroups. The dataset is organized into a semantic hierarchy of 6 different supercategories [2]. Some categories share close semantic proximity (e.g., comp.sys.ibm vs. comp.sys.mac), while others exhibit significant semantic divergence (e.g., sci.crypt vs. rec.sport). This hierarchical structure allows for granular anomaly detection scenarios, testing a model's ability to distinguish between subtle anomalies and obvious outliers.

For anomaly detection tasks, the dataset is typically adapted using a one-vs-rest protocol [22,18,5]. In this setup, one newsgroup is designated as the 'normal' class, while samples from the remaining 19 newsgroups are treated as anomalies. This creates a realistic 'needle-in-a-haystack' scenario where the model must learn the specific manifold of a single topic without supervision from the anomalous classes.

The AG News dataset is a large-scale news classification corpus constructed from the original AG's Corpus of News Articles [28]. It serves as a complementary benchmark to 20 Newsgroups, offering structured, professionally written text with lower noise levels than Usenet forum posts. The dataset contains 127,600 samples, split into 120,000 training and 7,600 testing samples. Each sample consists of the title and description fields of a news article. The data is categorized into four broad, semantically distinct classes: World, Sports, Business and Science.

The semantic distance between these four categories is relatively high (e.g., the vocabulary of 'Sports' is vastly different from 'Business'). This dataset is frequently used to benchmark semantic violation detection, where a model trained on 'Business' news must identify 'Sports' articles as distributional anomalies [22,18,5]. The high volume of training data per class (30,000 samples) allows for the training of deep generative models (like Llama-3 or GPT-Neo) without the risk of rapid overfitting often seen in smaller datasets.

[1] http://qwone.com/~jason/20Newsgroups/

[2] Computers, Miscellaneous, Politics, Recreation, Religion , Science.

### 3.2 Text representation strategies

The initial stage of our methodology transforms unstructured text into semantically rich vectors, which serve as the substrate for all downstream anomaly detection tasks. This process is foundational, as the quality of the learned representations directly constrains the potential performance of any subsequent algorithm. Contextual embeddings were chosen as a key focus as they generate dynamic, context-aware representations for each word instance, and the anomalous nature of a text often depends on subtle contextual cues. We use several variants of the BERT architecture [11], including RoBERTa [17](which optimizes the training process for improved robustness) and DeBERTa[12] (which uses disentangled attention to better model word positions and context). To address computational efficiency and scalability we also included 'lightweight' architectures such as DistilBERT[23], ALBERT[15], and ELECTRA [9]. Further, we include XLNet, an auto-regressive model that addresses certain limitations of BERT by introducing a permutation-based training objective [27].

To capture the capabilities of the most recent class of large-scale models, we also generate embeddings from large language models (LLMs) such as those from the GPT (generative pre-trained Transformer) family [7]. We employ GPT-2, DistilGPT-2 [23] and GPT-NEO [4] to assess how representations learned through predictive language modelling compare to bidirectional methods in flagging linguistic deviations. Lastly, we incorporate LLama-3 [2] and Gemini [24]. These represent larger-scaled pretrained models to test whether a superior baseline can help identify complex deviations.

### 3.3 Anomaly detection algorithms

Following the feature extraction stage, we then use a set of anomaly detection algorithms to identify deviations in the latent representations. These models span several methods. Neural-network-based approaches learn a compact representations of 'normality'. Autoencoders, variational autoencoders and contrastive autoencoders are employed to detect anomalies via high reconstruction error, under the premise that anomalous text will lack the common factors of variations found the data [14,20,21,1]. We also incorporate deep support vector data description (Deep SVDD) [21,20,22], which maps data into a minimum-volume hypersphere in a learned latent space, and context vector data description [21,20,22], a self-attentive model specifically designed to capture multiple semantic contexts within text corpora.

To evaluate how anomalies manifest in local data neighbourhoods, we apply Local Outlier Factor (LOF) [6] and DBSCAN [8]. LOF identifies outliers based on local density deviations compared to Neighbors while DBSCAN identifes sparse points that do not belong to dense clusters. Complementing these is Empirical Cumulative Distribution-based outlier Detection (ECOD) [26] a parameter-free statistical method that identifies "rare events" in the tails of the data distribution, offering high interpretability and computational efficiency.

Finally we investigate isolation-based and ensemble strategies. Isolation forest [1] is used to isolate anomalies through random attribute partitioning, leveraging the fact that outliers typically have shorter average path lengths in tree structures. To enhance stability and handle high-dimensional text features we also use feature bagging, which aggregates the scores of multiple base detectors. Lastly, we include a self-organizing GAN for anomaly detection (SOGAAL) [16] approach which uses a generative adversarial framework to sample informative regions of the feature space, improving the model's ability to distinguish between complex normal clusters and true anomalies.

### 3.4 Stand-alone models

In addition to the modular approach of pairing representation models with separate detection algorithms we also evaluate several integrated end-to-end frameworks. These models are designed to directly process raw text and learn to identify anomalies within a single unified architecture, often leveraging task-specific pre-training objectives. Their inclusion is crucial for comparing the performance of modular versus self-contained systems.

DATE [18] identifies anomalies by learning the intricate patterns of normal language through self-supervision. Through objectives such as replaced token detection, the model learns to distinguish between original and plausible but artificially replaced words. The process compels the architecture to construct a deep implicit model of semantic and syntactic regularities. DATE is included as a leading example of a purely self-supervised framework, testing the efficacy of detecting anomalies without exposure to any labelled instances.

FATE [10] is a deep learning model engineered for few-shot anomaly detection, addressing the practical challenge of data scarcity. The core mechanism involves deviation learning, where the model is explicitly trained to align normal representations with a reference score while pushing anomalous examples to deviate significantly. By leveraging multiple instance learning and multi-head self-attention, FATE can identify salient elements within documents.

## 4 Results and Analysis

In this section we present the evaluation of 14 pre-trained language models paired with 11 unsupervised anomaly detection algorithms, alongside 2 standalone frameworks. By benchmarking these 134 configurations across 10 distinct linguistic contexts (6 categories of the 20 Newsgroups dataset and the 4 categories of the AG news dataset), we aim to systematically decouple the impact of semantic representation from algorithmic detection logic, identifying the optimal strategies for context-aware anomaly detection.

Figure 1 visualizes the average F1 score of each combination of representation and model across the 10 different contexts, with the X-axis representing the representation layer and the Y-axis representing the detection layer. The heatmap exhibits horizontal stratification: The performance is dictated almost entirely by the anomaly detection method, while the choice of representation has a marginal effect. This suggests that the topological structure of anomalies in the embedding space is accessible to all tested Transformers, but only specific algorithms are capable of defining the boundary. The stand-alone models DATE and FATE also demonstrate strong performance with average F1 scores of 0.905 and 0.908 respectively across the same contexts.

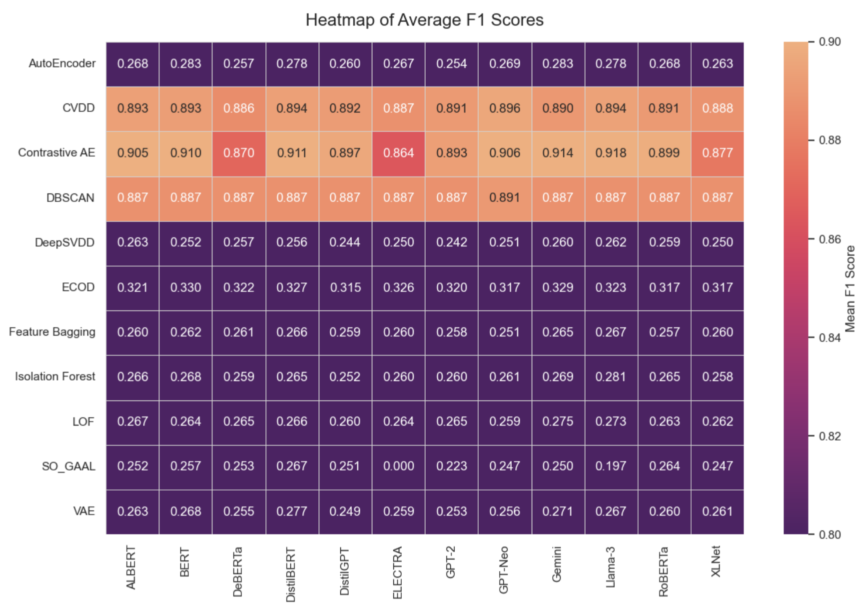


**Fig. 1.** Anomaly detection model performance across text representations; F1 scores averaged over 10 linguistic contexts (excluding standalone models).

Table 1 presents a breakdown of metrics for the top ten models as ranked by F1 score. The list is populated by deep anomaly detection methods, specifically contrastive autoencoders (7 out of 10 top spots) and DATE, along with FATE. More traditional methods like isolation forests and DBSCAN are entirely absent from the top tier. There is somewhat more variation in top-ranked representations. The superior performance of Llama-3 and Gemini suggests that the depth of semantic understanding required to detect subtle linguistic anomalies scales with model complexity. Llama-3 achieves a remarkable AUC of 0.945, significantly outperforming traditional encoders. This indicates that modern generative embeddings provide a cleaner separation between normality and abnormality.

Despite the dominance of massive LLMs, DistilBert ranks third, outperforming its own teacher (BERT) and larger peers like GPT-Neo and RoBERTa. This finding demonstrates that the core semantic features required to identify out-of-distribution text can be effectively captured and retained by compact, distilled

**Table 1.** AUC, precision, and recall figures averaged over all 10 contexts, for the top 10 models ranked by averaged F1 score.

| Language Model | AD Model | AUC | Precision | Recall | F1 Score |
|---|---|---|---|---|---|
| Llama-3 | Contrastive AE | 0.945 | 0.87 | 0.976 | 0.918 |
| Gemini | Contrastive AE | 0.902 | 0.857 | 0.98 | 0.913 |
| DistilBERT | Contrastive AE | 0.867 | 0.854 | 0.979 | 0.911 |
| BERT | Contrastive AE | 0.855 | 0.856 | 0.973 | 0.909 |
| FATE | FATE | 0.867 | 0.862 | 0.962 | 0.907 |
| GPT-Neo | Contrastive AE | 0.846 | 0.839 | 0.986 | 0.906 |
| ALBERT | Contrastive AE | 0.864 | 0.843 | 0.98 | 0.905 |
| DATE | DATE | 0.761 | 0.852 | 0.966 | 0.905 |
| RoBERTa | Contrastive AE | 0.794 | 0.826 | 0.986 | 0.899 |
| DistilGPT | Contrastive AE | 0.814 | 0.829 | 0.977 | 0.896 |

encoder architectures, without requiring the massive parameter space of generative LLMs. While CVDD achieves the highest recall in the entire benchmark, its overall rank drops due to lower precision and is not present in the top 10 combinations. This highlights a distinct architectural behaviour that CVDD functions as a 'hard' safety net that prioritizes safety over purity, whereas contrastive AE offers a more balanced decision boundary.

### 4.1 Comparative performance

In an unsupervised anomaly detection task a model that achieves high score on one dataset but fails on another is deemed unreliable. In this section we isolate the maximum F1 score achieved in each context, treating each dataset as an independent optimization problem. Through this, we identify the specific combination of language model and anomaly detection algorithm that captures the semantic topology of that specific domain best. Contrary to the above analysis of general performance, here we measure specialization.

**Table 2.** Best performing model per specific context.

| Context | Language Model | AD Model | F1 Score |
|---|---|---|---|
| 20ng_od_comp | BERT | Contrastive AE | 0.889 |
| 20ng_od_misc | Gemini | Contrastive AE | 0.980 |
| 20ng_od_pol | FATE | FATE | 0.936 |
| 20ng_od_rec | Llama-3 | Contrastive AE | 0.899 |
| 20ng_od_rel | BERT | Contrastive AE | 0.936 |
| 20ng_od_sci | Llama-3 | Contrastive AE | 0.889 |
| ag_od_business | Llama-3 | Contrastive AE | 0.904 |
| ag_od_sci | Llama-3 | Contrastive AE | 0.932 |
| ag_od_sports | FATE | FATE | 0.973 |
| ag_od_world | Gemini | Contrastive AE | 0.906 |

Table 2 suggests a dichotomy. On the one hand the anomaly detection method is almost static, with contrastive AE outperforming the other algorithms in 8 of 10 contexts, while the language model fluctuates significantly depending on the context. Llama-3 emerges as the clear winner in knowledge-intensive domains such as science, business and recreation. This suggests that recent LLMs possess superior domain knowledge when compared to older encoders. Despite the generative takeover, BERT appears more effective in domains characterised by dense and structurally rigid text, like religion and computing data. FATE excels in the most semantically distinct categories, politics and sports, while Gemini has the highest scores in the categories with the most diverse and least structured patterns.

### 4.2 Stability analysis

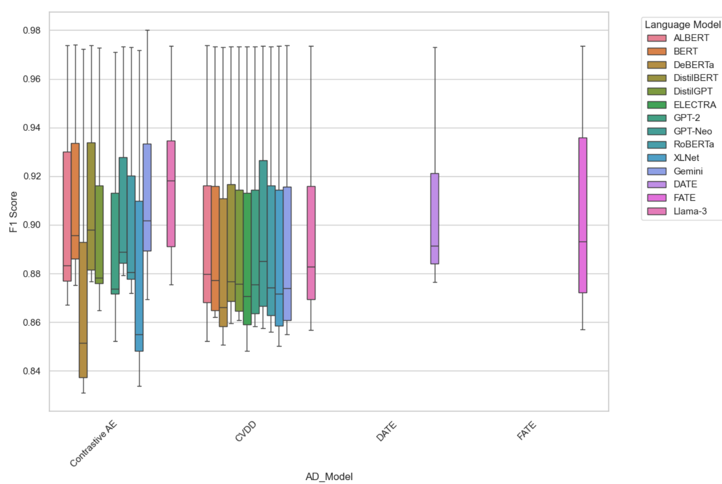


**Fig. 2.** Distribution of F1 scores per model and representation.

Figure 2 visualises the distribution of F1 scores across the 10 linguistic contexts. This figure isolates the top performers to evaluate their stability and context-awareness. A high variance indicates that a model is brittle, performing well on some datasets but poorly on others. However, extremely low variance hints to model collapse where the detection combination predicts normality for every data point. The CVDD cluster exhibits the highest degree of robustness. The distribution of F1 scores remains almost uniform in across all 12 language models characterised by compact IQRs and consistent medians. Contrastive AE achieves some of the highest individual medians, it displays significantly greater variance than CVDD. This indicates that Contrastive AE is more sensitive to the semantic qualities of the underlying embeddings.

DeBERTa exhibits the largest range and the longest whiskers fluctuating from approximately 0.83 to 0.98. While it excels at certain contexts it lacks the cross-domain stability seen in simpler distilled encoders like DistilBERT. FATE has the second longest whiskers in the entire comparison, spanning a range from F1 approximately 0.86 to 0.97. This confirms the findings above that FATE adapts well to favourable domains but lacks the stability of the other methods, resulting in significant volatility.

### 4.3 Contextual Vulnerability

We calculate a *performance delta* as the difference between a model's F1 Score on a specific target dataset versus its average F1 score across all datasets. A negative delta indicates a linguistic blind spot.

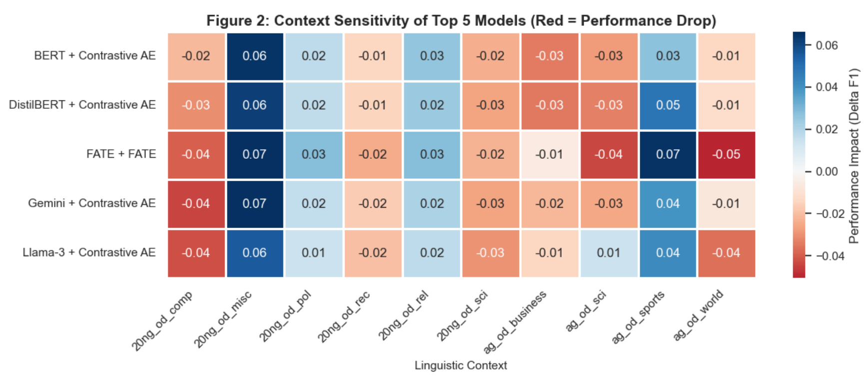


**Fig. 3.** Performance delta of the top 5 combinations.

Figure 3 visualises the performance delta for the top 5 model combinations. Blue highlights contexts where a model combination performed better than its average, while red indicates the opposite. With the worst drop being only $-0.04$ we find that the contrastive learning architecture is robust. The lack of deep red cells confirms that contrastive learning successfully creates a 'context-agnostic' latent space, in which the definition of an anomaly remains stable despite changing contexts. FATE exhibits the broadest range in performance (0.07 improvement and $-0.05$ drop) across the top 5 combinations which confirms the higher variance of the deviation based model compared to the other high performing combinations. It is worth mentioning that the Miscellaneous and Sports categories have better performance across all the combinations while the Computer and World News categories show the biggest drops.

## 5 Discussion

A prevailing assumption in unsupervised anomaly detection literature has been that performance gains are primarily driven by the quality of the representation layer rather than algorithmic processing in the detection layer. Our results provide empirical evidence to the contrary. Our context-specific analysis reveals a fundamental divergence in how different architectures model linguistic anomalies, delineated by the entropy of the domain. The consistent performance of contrastive autoencoders across diverse datasets validates the efficacy of metric learning. By optimizing for a latent distance metric rather than reconstruction error, contrastive AE creates a 'context-invariant' decision boundary. This architecture proves superior in stable, structured environments, effectively mitigating the noise often introduced by high-variance vocabulary. Conversely, the superior performance of FATE in high-entropy domains like politics and sports suggests that static manifolds are insufficient for rapidly evolving topics. In these contexts, the definition of "normal" is transient. FATE's success implies that deviation learning—calculating the relative distance of a sample from a local, temporal support set—is the optimal strategy for dynamic data streams, where the global distribution is non-stationary.

The emergence of Llama-3 and Gemini at the apex of the global ranking demonstrates that the depth of semantic understanding in the embedding space is a critical determinant of separability. While generative LLMs define the state-of-the-art, the performance of DistilBERT presents a significant finding for scalable deployment. Ranking third globally and outperforming significantly larger models, DistilBERT challenges the scaling laws typically associated with NLP tasks.

## 6 Conclusion

This study presents a comprehensive benchmarking of Transformer-based architectures for unsupervised linguistic anomaly detection. We established that manifold learning and deviation learning significantly outperform traditional reconstruction-based and density-based methods in high-dimensional semantic spaces. We provided evidence that integrating state-of-the-art generative LLMs as representation layers yields superior detection capabilities in knowledge-heavy domains. The results suggest that future research should move beyond the "one-size-fits-all" approach to anomaly detection. The distinct success of FATE in dynamic domains versus contrastive AE in static domains argues for adaptive meta-learning frameworks that can dynamically select the detection strategy based on the entropy of the input stream. Ultimately, this work demonstrates that unsupervised linguistic anomaly detection has evolved from a purely statistical exercise into a semantic reasoning task. As textual data becomes increasingly sophisticated, our detection mechanisms must transition from identifying statistical outliers to understanding semantic deviations. The architectures identified in this paper provide a foundational blueprint for the next generation of robust, scalable content moderation systems.

Future work will expand upon this foundational methodology by transitioning from document-level classification to entity-centric anomaly detection. This transition requires surveying model performance across a significantly wider variety of online content. Doing so will allow us to investigate whether the optimal architectural combinations identified in this study remain robust when isolating targeted, entity-specific semantic deviations across diverse domains.

## Acknowledgments

This research is made possible through the UK EPSRC [EP/S022465/1] and Isambard-AI resources provided by AIRR, funded by DSIT via UKRI and STFC [ST/AIRR/I-A-I/1023].

## References

1. Aggarwal, C.C., Aggarwal, C.C.: An introduction to outlier analysis. Springer (2017)
2. AI@Meta: Llama 3 model card (2024), https://github.com/meta-llama/llama3/blob/main/MODEL_CARD.md
3. Albishre, K., Albathan, M., Li, Y.: Effective 20 newsgroups dataset cleaning. In: 2015 IEEE/WIC/ACM International Conference on Web Intelligence and Intelligent Agent Technology (WI-IAT). vol. 3, pp. 98–101. IEEE (2015)
4. Black, S., Gao, L., Wang, P., Leahy, C., Biderman, S.: GPT-Neo: Large scale autoregressive language modeling with mesh-tensorflow (2021), http://github.com/eleutherai/gpt-neo
5. Breidenstein, A., Labeau, M.: Using locally learnt word representations for better textual anomaly detection. In: Proceedings of the Fifth Workshop on Insights from Negative Results in NLP. pp. 82–91 (2024)
6. Breunig, M.M., Kriegel, H.P., Ng, R.T., Sander, J.: Lof: identifying density-based local outliers. In: Proceedings of the 2000 ACM SIGMOD international conference on Management of data. pp. 93–104 (2000)
7. Brown, T., Mann, B., Ryder, N., Subbiah, M., Kaplan, J.D., Dhariwal, P., Neelakantan, A., Shyam, P., Sastry, G., Askell, A., et al.: Language models are few-shot learners. Advances in neural information processing systems **33**, 1877–1901 (2020)
8. Chandola, V., Banerjee, A., Kumar, V.: Anomaly detection: A survey. ACM computing surveys (CSUR) **41**(3), 1–58 (2009)
9. Clark, K., Luong, M.T., Le, Q.V., Manning, C.D.: ELECTRA: Pre-training text encoders as discriminators rather than generators. In: ICLR (2020), https://openreview.net/pdf?id=r1xMH1BtvB
10. Das, A.S., Ajay, A., Saha, S., Bhuyan, M.: Few-shot anomaly detection in text with deviation learning. In: International Conference on Neural Information Processing. pp. 425–438. Springer (2023)
11. Devlin, J., Chang, M.W., Lee, K., Toutanova, K.: Bert: Pre-training of deep bidirectional transformers for language understanding. In: Proceedings of the 2019 conference of the North American chapter of the association for computational linguistics: human language technologies, volume 1 (long and short papers). pp. 4171–4186 (2019)

12. He, P., Liu, X., Gao, J., Chen, W.: Deberta: Decoding-enhanced bert with disentangled attention. In: International Conference on Learning Representations (2021), https://openreview.net/forum?id=XPZIaotutsD
13. Islam, M.R., Liu, S., Wang, X., Xu, G.: Deep learning for misinformation detection on online social networks: a survey and new perspectives. Social Network Analysis and Mining **10**(1), 82 (2020)
14. Jafari, A.: A deep learning anomaly detection method in textual data. arXiv preprint arXiv:2211.13900 (2022)
15. Lan, Z., Chen, M., Goodman, S., Gimpel, K., Sharma, P., Soricut, R.: ALBERT: A lite BERT for self-supervised learning of language representations. CoRR **abs/1909.11942** (2019), http://arxiv.org/abs/1909.11942
16. Liu, Y., Li, Z., Zhou, C., Jiang, Y., Sun, J., Wang, M., He, X.: Generative adversarial active learning for unsupervised outlier detection. IEEE Transactions on Knowledge and Data Engineering **32**(8), 1517–1528 (2019)
17. Liu, Y., Ott, M., Goyal, N., Du, J., Joshi, M., Chen, D., Levy, O., Lewis, M., Zettlemoyer, L., Stoyanov, V.: Roberta: A robustly optimized bert pretraining approach. arXiv preprint arXiv:1907.11692 (2019)
18. Manolache, A., Brad, F., Burceanu, E.: Date: Detecting anomalies in text via self-supervision of transformers. arXiv preprint arXiv:2104.05591 (2021)
19. Rao, S., Verma, A.K., Bhatia, T.: A review on social spam detection: Challenges, open issues, and future directions. Expert Systems with Applications **186**, 115742 (2021)
20. Ruff, L., Kauffmann, J.R., Vandermeulen, R.A., Montavon, G., Samek, W., Kloft, M., Dietterich, T.G., Müller, K.R.: A unifying review of deep and shallow anomaly detection. Proceedings of the IEEE **109**(5), 756–795 (2021)
21. Ruff, L., Vandermeulen, R., Goernitz, N., Deecke, L., Siddiqui, S.A., Binder, A., Müller, E., Kloft, M.: Deep one-class classification. In: Dy, J., Krause, A. (eds.) Proceedings of the 35th International Conference on Machine Learning. Proceedings of Machine Learning Research, vol. 80, pp. 4393–4402. PMLR (10–15 Jul 2018), https://proceedings.mlr.press/v80/ruff18a.html
22. Ruff, L., Zemlyanskiy, Y., Vandermeulen, R., Schnake, T., Kloft, M.: Self-attentive, multi-context one-class classification for unsupervised anomaly detection on text. In: Korhonen, A., Traum, D., Màrquez, L. (eds.) Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics. pp. 4061–4071. Association for Computational Linguistics, Florence, Italy (Jul 2019). https://doi.org/10.18653/v1/P19-1398, https://aclanthology.org/P19-1398/
23. Sanh, V., Debut, L., Chaumond, J., Wolf, T.: Distilbert, a distilled version of bert: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108 (2019)
24. Team, G., Anil, R., Borgeaud, S., Alayrac, J.B., Yu, J., Soricut, R., Schalkwyk, J., Dai, A.M., Hauth, A., Millican, K., et al.: Gemini: a family of highly capable multimodal models. arXiv preprint arXiv:2312.11805 (2023)
25. Xiao, F., Fan, J.: Text-adbench: Text anomaly detection benchmark based on llms embedding. arXiv preprint arXiv:2507.12295 (2025)
26. Yang, T., Nian, Y., Li, S., Xu, R., Li, Y., Li, J., Xiao, Z., Hu, X., Rossi, R., Ding, K., et al.: Ad-llm: Benchmarking large language models for anomaly detection. arXiv preprint arXiv:2412.11142 (2024)
27. Yang, Z., Dai, Z., Yang, Y., Carbonell, J., Salakhutdinov, R.R., Le, Q.V.: Xlnet: Generalized autoregressive pretraining for language understanding. Advances in neural information processing systems **32** (2019)
28. Zhang, X., Zhao, J., LeCun, Y.: Character-level convolutional networks for text classification. Advances in neural information processing systems **28** (2015)